

Chemical Kinetics, Reaction Coordinates and Temperature Dependence

Department of Physical Chemistry - Chemical Dynamics Lecture Series

1. Rate Laws and Reaction Order

For reaction $aA + bB \rightarrow cC + dD$, the experimental rate law takes the general form:

$$\text{Rate} = k * [A]^m * [B]^n$$

The reaction orders m and n must be determined empirically and cannot be inferred from stoichiometry.

2. The Arrhenius Equation

The temperature dependence of the reaction rate constant k is modeled by the Arrhenius equation:

$$k = A * \exp(-E_a / (R * T))$$

where A is the pre-exponential frequency factor, E_a is the activation energy in J/mol,

R is the universal gas constant (8.314 J/(mol K)), and T is absolute temperature in Kelvin.

Linearization of the Arrhenius relation yields the standard diagnostic plot:

$$\ln(k) = \ln(A) - (E_a / R) * (1 / T)$$

Plotting $\ln(k)$ vs $1/T$ produces a straight line whose slope is $-E_a / R$.

Table 1: Kinetic rate constants for ester hydrolysis across temperature range.

Temperature (K) | $1/T$ (10^{-3} K^{-1}) | k ($\text{M}^{-1} \text{ s}^{-1}$) | $\ln(k)$

283.15 | 3.532 | 0.00142 | -6.557

298.15 | 3.354 | 0.00512 | -5.275

313.15 | 3.193 | 0.01680 | -4.086

328.15 | 3.047 | 0.05040 | -2.988

Figure 1 (Page 1): Arrhenius plot of $\ln(k)$ versus inverse temperature showing linear activation slope.